Unemployment in India & The Periodic Labour Force Survey

Final Project Report for **Group 'B'**

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1 Introduction

The *Periodic Labour Force Survey* is a quarterly survey conducted every year by the National Statistical Office (NSO) (under the **Ministry of Statistics and Programme Implementation**, **Government of India**) to release estimates of labour force indicators, such as *Labour Force Participation Rate (LFPR)*, *Worker Population Ration (WPR)*, *Unemployment Rate (UR)*, distribution of workers by broad status in employment and industry of work in the *Current Weekly Status (CWS)* for the **urban** areas. These surveys are the primary source of statistics on labour force, activity participation of the population and structure of employment and unemployment in the country. The architecture of these surveys provides the measurement of labour force indicators in cross classification of age, gender, education, industry, occupation, time disposition, mobility and wages.

From 2019 onwards (with the pilot sample being launched in 2018), a rotational panel sampling design has been incorporated in the urban areas. In this rotational panel scheme, each selected household in the urban areas is visited four times: in the beginning with the '*First Visit Schedule*' and thrice periodically later, with a '*Revisit Schedule*'. The scheme of rotation ensures that **75%** of the first-stage sampling units (**FSU**s) are matched between two consecutive units. This is necessary in order to compare several key indicators among same households in differing quarters.

As the units in different quarters are *not independent*, this sampling design poses a problem when we want to test whether there is some significant difference, with respect to some indicator, in two different quarters, using our known tests, which require the assumption of independence. In this report, we have *proposed a testing procedure* to check if indeed such a difference is present between the quarters. Unemployment rate is one of the crucial indicators, due to the obvious fact that a significant rise in unemployment is concerning for the economy and reasons for such a change must be identified and dealt with.

We have mainly focused on identifying differences in Unemployment Rate and Monthly Consumer Expenditure of households, though the methodology can easily be extended to other factors (the same statistic can be used). We have tested differences for each quarter based on gender, age, educational qualifications as well as the total data on different States as well as over entire India. Some figures on UR for small states (less population) have not been reported in the original survey and thus we have excluded them. Testing for all major States have been done. Finally, we have also tested differences between expenditure of households in different quarters, in order to implement the continuous counterpart of our test statistic.

2 Definitions

We first define the necessary terms to avoid any ambiguity in understanding. The detailed descriptions of concepts and definitions relevant to the survey can be found here[1].

1. **Activity Status- Current Weekly Status (CWS)**: The activity status determined on the basis of a reference period of last 7 days preceding the date of survey is known as the current weekly status (CWS) of the person.

   a) **Principal Activity Status (PAS)**- The activity status on which a person spent relatively long time (major time criterion) during 365 days preceding the date of survey, was considered the usual principal activity status of the person.

   b) **Subsidiary Activity Status (SAS)**- The activity status in which a person in addition to his/her usual principal status, performs some economic activity for 30 days or more for the reference period of 365 days preceding the date of survey, was considered the subsidiary economic status of the person.

2. **Unemployment Rate (UR)**: UR is defined as the percentage of persons unemployed among the persons in the labour force. The people are categorised into three sections: *Employed (E)*, *Unemployed (U)* and *Not in Labour Force (O)*.

Whether a person is unemployed or not can be determined from his/her CWS. If the person has not worked for even 1 hour on any day during the reference week but sought or was available for work at least for 1 hour on any day during the reference week.[2] If the Principal Activity Status of the person is recorded as '81' or '82', then the person is classified as unemployed[1]. Then UR is defined as:

$$\text{UR} = \frac{U}{E + U}$$

3 Sampling Method and Data

3.1 Sampling Method:

As explained before, a rotational sampling scheme is employed in the urban. The rotational scheme will be of two years duration to accommodate the changes in the urban frame in the intracensal period; in the sense that the sampling frames for both rural and urban areas will remain unchanged for every two-year duration. In this rotational panel scheme each selected household in urban areas will be visited four times – one with first visit schedule and other three with revisit schedule. The estimates can be given for successive quarters without any break in the series (starting from the fifth quarter), ensuring a 75% matching between consecutive quarters.

Regression based estimates will not be generated. Instead, usual traditional design based estimates will be generated. The proposed design aims at generating quarterly estimates of level and change parameters of some important labour force indicators (LFPR, WPR & UR) based on CWS data in urban areas and annual estimates of level parameters based on usual status for both rural and urban areas in the line of employment & unemployment survey of NSS quinquennial round. For more details, see [1].

For rural areas, samples for all the 8 quarters will be selected before commencement of survey for each two-year period, while the frame remains same for this duration. In each quarter, only 25% FSUs of annual allocation (as is done in each sub-round of NSS rounds) will be covered in rural areas so that independent estimates can be generated for each quarter. There will not be any revisit in the rural samples.

3.2 Data:

The number of FSUs (urban households) that were actually visited in the various quarters (for the first, second, third or fourth time) are presented in the table below:

Table 1: Number of urban households visited

Visit / Quarter	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Visit 1	11502	11526	11018	11143
Visit 2	10986	11427	11235	10609
Visit 3	11033	11238	11070	10838
Visit 4	10954	11430	10648	10679

4 Summary and Key Findings

The following observations have been made using the testing procedure for the PLFS Data. The detailed tables have been provided later. HCE represents Household Consumption Expenditure.

Quarters	Unemployment		HCE	
	T	p-value	T	p-value
Q1 vs Q2	1.1048	0.26923	-4.627	0.0003
Q2 vs Q3	-10.149	<0.0001	-5.23	<0.0001
Q3 vs Q4	-61.46	<0.0001	-2.334	0.019

1. For the whole data (no divisions), significant difference was observed between the Quarters 2 and 3 (p-value < 0.0001) and for Quarter 3 and 4 (p-value < 0.0001). This, can be well explained due to the country wide lockdown that was present, during this period (starting March, 2020) which led to the collapse of several industries and hence the rise in unemployment.
2. Across almost all the states, this effect of lockdown was noticed, which is evident from the high values of the T statistic between Quarters 3 and 4.
3. Across gender, the above trend was present for males but less so for females (across states), which suggest that the overall female working population did not change much in many of the states.
4. In terms of education, a significant difference was observed between Quarters 3 and 4, for the graduate and post-graduate students, which is expected, as many lost their jobs in industries. However, no significant difference was obtained in any of the other quarters.
5. The Consumer Expenditure study provided an erratic pattern which might be attributed to such a volatile economy during that period. In quarters 3 and 4, the difference was *not* significant, suggesting that the households, having grown accustomed to the lockdown, kept track of their monthly expenditure, in case of emergencies.

In any case, 2019-2020 presented as a rare event in which most of the differences can be attributed to the breakout of the Corona virus disease and the implementation of the lockdown.

5 Testing Procedure and Methodology

We consider a very general setup. Consider two consecutive quarters Q_a and $Q_{(a \mod 4)+1}$ during a particular survey period. Suppose S_1 and S_2 denote the sample units surveyed in Q_a and $Q_{(a \mod 4)+1}$ respectively. Let

$$\mathbf{S_1} = \{X_1,\ X_2, \dots\ X_{m_1},\ C_1,\ C_2 \dots,\ C_n\}\ ;\ \mathbf{S_2} = \{Y_1,\ Y_2, \dots\ Y_{m_2},\ C_1,\ C_2 \dots,\ C_n\}$$

with the condition: $\{X_1,\ X_2, \dots\ X_{m_1},\ C_1,\ C_2 \dots,\ C_n\} \cap \{Y_1,\ Y_2, \dots\ Y_{m_2}\} = \phi$

Thus, X_i's and Y_i's represent the independent units and C_i's represent the common units between the two quarters. For recording the responses, it is convenient to use the same symbols. Thus, let $\{X_i\}_{i=1}^{m_1}$ and $\{Y_i\}_{i=1}^{m_2}$ be the responses from the independent units and $\{C_{1i}\}_{i=1}^{n}$ and $\{C_{2i}\}_{i=1}^{n}$ be the responses from the dependent units in Q_a and $Q_{(a \mod 4)+1}$. respectively.

We wish to check if there is a significant amount of difference between the recorded values in $\mathbf{S_1}$ and $\mathbf{S_2}$.

5.1 Hypothesis:

We test for the simplest hypothesis of equality of means.

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$$

5.2 Assumptions:

1. $X_1,\ X_2, \dots\ X_{m_1},\ C_{11},\ C_{12} \dots,\ C_{1n} \sim\ F(\mu_1,\ \sigma^2)$
2. $Y_1,\ Y_2, \dots\ Y_{m_2},\ C_{21},\ C_{22} \dots,\ C_{2n} \sim\ F(\mu_2,\ \sigma^2)$

We assume that the distribution function of the data in both the samples have the *same functional form (only differing in the population mean)*. This is a reasonable assumption as we do not expect the distribution (the data generation procedure) to change all of a sudden when we survey the next quarter. The mean might change with time and this is the only difference that we are interested in.

The assumption on equality of variance stems from the same belief that there is no reason to assume that the variation in such macro indicators change drastically in such a small span of time (between two quarters, i.e, within 3 months). However, this assumption *can be dropped* (using Welch's t-Test) and the procedure would still work. We work with the assumption of equal variances to make our statistic simpler.

We *do not employ* a two step procedure of first checking equality of variances (using Levene's Test) and then checking for equality of means, as this has been proven to be a worse procedure [3].

5.3 Problems with Present Tests:

We begin with the most simple tests and state the problems that occur.

1. We first start with Parametric Tests. Of course, the Normal t-test for difference of two means cannot be employed as the observations in the two samples are **not independent**. Also, the assumption of normality is often not justified.

   Most of the indicators have a highly skewed distribution and attempts to make them normal (using popular transformations like: Power Transformations such as, Box-Cox and Yeo-Johnson transformations, Ordered Quantile Normalization transforms, etc.) have yielded infeasible results. Thus, developing a statistic, assuming that the data is normal, and deriving properties based on it, might result in inaccurate results.

   As a large number of indicators, with varying distributions need to be taken into account, a parametric test seems to be too *constrained* and unreasonable for our purposes.

2. Next we move to Non-parametric Tests. Equality of two distributions can be checked using a variety of tests (*Two Sample K-S* test, *Mann-Whitney* Test, *Wilcoxon Signed Rank* Test, etc. (ref)). Both the K-S Test and M-W Test need independent samples and can be employed for the independent counterparts ($\mathbf{X}$ and $\mathbf{Y}$). However, they cannot be used for testing difference among the dependent parts ($\mathbf{C_1}$ and $\mathbf{C_2}$). For this purpose, the Wilcoxon Signed Rank Test can be used to test for the difference in this paired sample.

   Thus, we divide our samples into two parts, the independent part ($\mathbf{X}$ and $\mathbf{Y}$) and the dependent part ($\mathbf{C_1}$ and $\mathbf{C_2}$). Now, for a given level of significance (say, α), we are testing these two parts *separately* (the independent part using **M-W Test**, and the dependent part using **Wilcoxon Signed Rank Test**). Thus, we need to have a multiple testing procedure with different level of significance for the two tests, such that the overall level does not overshoot α. However, this procedure has the following disadvantages:

   (a) **Weightage:** Simple techniques like Bonferroni's Method suggest testing these two parts at $\alpha/2$ level separately and rejecting the hypothesis of equality of means if any of them is violated. However, is giving *equal weights* (in terms of level of significance) for both the dependent and independent parts justified? It seems that the dependent part gives more information as to the actual change in means, as the observations are recorded for the same units over time. Thus, the test for dependent parts should receive more weightage (even a small difference among the dependent parts could reflect actual significant change which the independent part might not).

However it is difficult to assign any such weight based on some procedure. For this reason, multiple testing seems to be a less appealing option for this study.

(b) **Confidence Interval:** When we are employing this multiple testing procedure, we would not be able to get a *single* confidence interval for the difference of two means (we can get two confidence intervals individually, but then it is difficult to incorporate them into a single C.I.)

(c) **Discrete Data**: In certain cases, like unemployment, the data obtained is *discrete* and it is not possible to apply such non-parametric methods which need the assumption of a continuous distribution function.

(d) **Entire data is not exploited:** We do not look at interactions or changes between $\mathbf{X}$ and $\mathbf{C_2}$ or $\mathbf{Y}$ and $\mathbf{C_1}$ in this procedure, which could lead to a possible loss of information.

For these reasons, we seek to provide a *single statistic* that does not suffer from the disadvantages of multiple testing and whose asymptotic distribution can be easily found out. We assume that $\text{Corr}(\mathbf{C_1},\ \mathbf{C_2}) = \rho$.

5.4 Test Statistic:

The proposed test statistic for checking difference in means is:

$$\boxed{T = \frac{1}{\sqrt{2}\hat{\sigma}}\left[\sqrt{\frac{m_1 m_2}{m_1+m_2}}(\bar{\mathbf{X}} - \bar{\mathbf{Y}}) + \sqrt{\frac{n}{2(1-\hat{\rho})}}(\bar{\mathbf{C}}_\mathbf{1} - \bar{\mathbf{C}}_\mathbf{2})\right]}$$

where,

(i) $\bar{\mathbf{X}} = \frac{1}{m_1}\sum_{i=1}^{m_1} X_i\ ,\ \ \bar{\mathbf{Y}} = \frac{1}{m_2}\sum_{i=1}^{m_2} Y_i$

(ii) $\bar{\mathbf{C}}_\mathbf{i} = \frac{1}{n}\sum_{j=1}^{n} C_{ij}\ ,\ \ i \in [2]$

(iii) $\hat{\sigma}^2 = \frac{1}{m_1+m_2+n-3}\left[\sum_{i=1}^{m_1}(X_i - \bar{\mathbf{X}})^2 + \sum_{i=1}^{m_2}(Y_i - \bar{\mathbf{Y}})^2 + \sum_{i=1}^{n}(C_{1i} - \bar{\mathbf{C}}_\mathbf{1})^2\right]$

(iv) $\hat{\rho} = \hat{\text{Corr}}(\mathbf{C_1},\ \mathbf{C_2}) = \dfrac{\sum_{i=1}^{n}(C_{1i} - \bar{\mathbf{C}}_\mathbf{1})(C_{2i} - \bar{\mathbf{C}}_\mathbf{2})}{n\hat{\sigma}^2}$

5.5 Distribution and Properties of Test Statistic:

Under our model assumptions, $\hat{\sigma}^2$ is an unbiased estimate of σ^2 and $\hat{\sigma}^2 \xrightarrow{\mathbb{P}} \sigma^2$. (click here) Using CLT and Slustky's Theorem, one can show the following results, under the hypothesis of equality of means. The proofs have been given provided below:

$$T_1 = \frac{1}{\hat{\sigma}}\sqrt{\frac{m_1 m_2}{m_1 + m_2}}(\bar{\mathbf{X}} - \bar{\mathbf{Y}}) \xrightarrow{d} \mathcal{N}(0,\ 1) \quad ; \quad T_2 = \frac{1}{\hat{\sigma}}\sqrt{\frac{n}{2(1-\hat{\rho})}}(\bar{\mathbf{C}}_{\mathbf{1}} - \bar{\mathbf{C}}_{\mathbf{2}}) \xrightarrow{d} \mathcal{N}(0,\ 1)$$

Now as $(\mathbf{X},\ \mathbf{Y})$ is independent of $(\mathbf{C_1},\ \mathbf{C_2})$, T_1 is independent of T_2. Thus,

$$T = \frac{1}{\sqrt{2}}(T_1 + T_2) \xrightarrow{d} \mathcal{N}(0,\ 1) \qquad \text{(Under } H_0\text{)}$$

Proof:

For T_1, as $\{X_i\}_{i=1}^{m_1}$ and $\{Y_i\}_{i=1}^{m_2}$ are i.i.d observations, and

$$T_1' = \frac{\bar{\mathbf{X}} - \mu_1 - (\bar{\mathbf{Y}} - \mu_2)}{\hat{\sigma}\sqrt{\frac{1}{m_1} + \frac{1}{m_2}}}$$

is simply a two sample t-statistic, which is asymptotically normal (click here). For H_0, T_1' reduces to T_1. Now the asymptotic normality of T_2 can also be shown using the same procedure. Now,

$$T_2' = \frac{\bar{\mathbf{C}}_{\mathbf{1}} - \mu_1 - (\bar{\mathbf{C}}_{\mathbf{2}} - \mu_2)}{\hat{\sigma}\sqrt{\frac{2(1-\hat{\rho})}{n}}}$$

Now, $\text{Var}(\mathbf{C_1} - \mathbf{C_2}) = 2\sigma^2(1-\rho)$. If we can show that $2\hat{\sigma}^2(1-\hat{\rho}) \xrightarrow{\mathbb{P}} 2\sigma^2(1-\rho)\ldots$ (1),

then again using Slustky's Theorem, the asymptotic normality of T_2' would follow, which reduces to T_2 under H_0. Now consider the sample variance of $\mathbf{C_1} - \mathbf{C_2}$

$$\begin{aligned}\hat{\text{Var}}(\mathbf{C_1} - \mathbf{C_2}) &= \frac{1}{n}\sum_{i=1}^{n}(C_{1i} - C_{2i})^2 - n(\bar{\mathbf{C}}_{\mathbf{1}} - \bar{\mathbf{C}}_{\mathbf{2}})^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}[C_{1i}^2 + C_{2i}^2 - 2C_{1i}\cdot C_{2i}] - n(\bar{\mathbf{C}}_{\mathbf{1}} - \bar{\mathbf{C}}_{\mathbf{2}})^2\end{aligned}$$

$$= \frac{1}{n}\sum_{i=1}^{n} C_{1i}^2 - n\bar{\mathbf{C}}_{\mathbf{1}}^2 + \frac{1}{n}\sum_{i=1}^{n} C_{2i}^2 - n\bar{\mathbf{C}}_{\mathbf{2}}^2 - 2(\frac{1}{n}\sum_{i=1}^{n} C_{1i}\cdot C_{2i} - n\bar{\mathbf{C}}_{\mathbf{1}}\bar{\mathbf{C}}_{\mathbf{2}})$$
$$= \hat{\sigma_1}^2 + \hat{\sigma_2}^2 - 2\hat{\rho}\hat{\sigma_1}\hat{\sigma_2} \xrightarrow{\mathbb{P}} 2\sigma^2(1-\rho)$$

Now, as $\frac{\hat{\sigma_1}}{\hat{\sigma}} \xrightarrow{\mathbb{P}} 1$, $\frac{\hat{\sigma_2}}{\hat{\sigma}} \xrightarrow{\mathbb{P}} 1$ as $n \to \infty,\ m_2 \to \infty$, it is easy to see that $\hat{\rho} \xrightarrow{\mathbb{P}} \rho$ (can also be proved easily using WLLN if the asymptotic variance goes to zero, which does in this case). Now as $\hat{\sigma}^2 \xrightarrow{\mathbb{P}} \sigma^2$ as $m_1,\ m_2,\ n \to \infty$, eq. (1) holds.

5.6 Benefits over Multiple Testing:

Clearly, this test statistic does not face the problems due to multiple testing. T can also be viewed as a weighted sum of the two parts (dependent and independent) which we wanted in the first part:

$$T = w_1(\bar{\mathbf{X}} - \bar{\mathbf{Y}}) + w_2(\bar{\mathbf{C}}_{\mathbf{1}} - \bar{\mathbf{C}}_{\mathbf{2}}) \quad ;\ w_1 = \frac{1}{\hat{\sigma}}\sqrt{\frac{m_1 m_2}{2(m_1+m_2)}} \text{ and } w_2 = \frac{1}{2\hat{\sigma}}\sqrt{\frac{n}{(1-\hat{\rho})}}(\bar{\mathbf{C}}_{\mathbf{1}} - \bar{\mathbf{C}}_{\mathbf{2}})$$

In our study $m_1 \approx m_2$ and $n \approx 3m_1$ (75% common). Thus,

$$\frac{w_1}{w_2} \approx \sqrt{\frac{m^2}{4m}} \cdot \sqrt{\frac{4(1-\hat{\rho})}{3m}} = \sqrt{\frac{1-\hat{\rho}}{3}} \quad (m_1 \approx m_2 = m \text{ say})$$

Thus, if $\hat{\rho}$ is close to 1 (perfect linear dependence), then the independent part would receive very less weight. If $\hat{\rho}$ is close to 0, then the two parts receive wights based on the ratio of their sample sizes. In case the data is normally distributed, $\hat{\rho}$ close to 0 would indicate almost independent data and the two parts can be pulled together to give a common sample. The resulting t-distributed statistic in such a case would have the same form as our statistic, with the only difference that the common part would be inflated by a certain factor.

As a special case, if the dependent data is distributed as a Bivariate Normal Distribution, then β_{OLS} can also serve as an unbiased estimator of ρ in this setup, due to the equality of variances and its normality properties may be exploited. Also, in that case, if both $\mathbf{X}$ and $\mathbf{Y}$ are normal, then T would be a sum of two $t_{m_1+m_2+n-3}$ distributed random variables. However, this would be a very restrictive setup and we avoid its analysis for our current purposes.

For our testing purposes, we have reported the value of the statistic T, as well as the p-values.

5.7 Confidence Interval:

From our previous discussions, with w_1 and w_2 defined

$$\boxed{T \xrightarrow{d} \mathcal{N}\left((w_1 + w_2)(\mu_1 - \mu_2),\ 1\right)}$$

If $\tau_{\alpha/2}$ denotes the $\alpha/2$-th quantile of the $\mathcal{N}(0, 1)$ distribution, then,

$$\begin{aligned} & \mathbb{P}\left(-\tau_{\alpha/2} < T - (w_1 + w_2)(\mu_1 - \mu_2) \leq \tau_{\alpha_2}\right) \leq \alpha \\ \implies & \mathbb{P}\left(\mu_1 - \mu_2 \in \left[\frac{T - \tau_{\alpha/2}}{w_1 + w_2}, \frac{T + \tau_{\alpha/2}}{w_1 + w_2}\right]\right) \leq \alpha \end{aligned}$$

Thus, the $100(1 - \alpha)\%$ C.I. for difference in means is:

$$\boxed{\left[\frac{T - \tau_{\alpha/2}}{w_1 + w_2}, \frac{T + \tau_{\alpha/2}}{w_1 + w_2}\right]}$$

5.8 Statistic for Unemployment Study:

For Unemployment Data, we record an individual as either unemployed or not unemployed based on the codes specified. Thus, each X_i, Y_j and C_k are Bernoulli random variables. Thus we assume that:

$$X_i \overset{i.i.d}{\sim} \text{Ber}(p_1) \quad ; \quad Y_i \overset{i.i.d}{\sim} \text{Ber}(p_2)$$

Also, X_i and Y_j's are independent. Now for the dependent part, we assume that:

$$(\mathbf{C_1},\ \mathbf{C_2}) \sim \text{Bivariate Binomial}\ (n,\ p_1,\ p_2,\ \rho)$$

Here,

$$\rho = \frac{\pi_{11} - p_1 p_2}{\sqrt{p_1(1 - p_1)p_2(1 - p_2)}} \qquad \text{where, } \pi_{11} = \mathbb{P}(C_{1i} = 1,\ C_{2i} = 1)$$

The marginals are just $\text{Bin}(n,\ p_i)$ distributed[4], with the added feature that now they have a correlation of ρ. Now the same statistic T, as defined before, would also work for this purpose. The only thing is to find estimates for $\hat{\sigma}$ and $\hat{\rho}$.

(i) $\hat{p_1} = \frac{1}{n}\sum_{i=1}^{n} C_{1i} \quad ; \quad \hat{p_2} = \frac{1}{n}\sum_{i=1}^{n} C_{2i}$

(ii) $\hat{\pi}_{11} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(C_{1i} = 1,\ C_{2i} = 1)$

(iii) $\hat{\sigma}^2 = \hat{p}(1-\hat{p})$, where $\hat{p} = \frac{1}{m_1+m_2+n}(\sum_{i=1}^{m_1} X_i + \sum_{i=1}^{m_2} Y_i + \sum_{i=1}^{n} C_{1i})$

(iv) $\hat{\rho} = \hat{\text{Corr}}(\mathbf{C_1},\ \mathbf{C_2}) = \dfrac{\hat{\pi}_{11} - \hat{p_1}\hat{p_2}}{\sqrt{\hat{p_1}(1-\hat{p_1})\hat{p_2}(1-\hat{p_2})}}$

Now, it can be shown[4] that under H_0,

$$\left(\frac{\sqrt{n}(\hat{p}_1 - p_1)}{\hat{\sigma}},\ \frac{\sqrt{n}(\hat{p}_2 - p_2)}{\hat{\sigma}}\right) \sim \text{Bivariate Normal } (0,\ 0,\ 1,\ 1,\ \rho)$$

Hence,

$$\sqrt{n}(\hat{p}_1 - \hat{p}_2) \xrightarrow{\mathbb{P}} \mathcal{N}(0,\ 2\hat{\sigma}^2(1-\hat{\rho}))$$

Hence, the same statistic can be used with exactly the same distributional properties.

5.9 Lyapunov's CLT: The General Test Statistic

In our original problem, a stratified sampling method has been used and thus estimates for the indicators are a weighted average of the strata estimates and thus a simple straight average would not suffice (as we have done earlier). In such a case, Lyapunov's CLT or other similar forms maybe used to obtain the asymptotic distributions.

$$T = \left[w_1'(\mathbf{q_1'}\bar{\mathbf{X}} - \mathbf{q_2'}\bar{\mathbf{Y}}) + w_2'(\mathbf{r_1'}\bar{\mathbf{C}}_\mathbf{1} - \mathbf{r_2'}\bar{\mathbf{C}}_\mathbf{2})\right]$$

where,

(i) $\mathbf{q_1'}\bar{\mathbf{X}} = \dfrac{\sum_{i=1}^{m_1} q_{1i}X_i}{\sum_{i=1}^{m_1} q_{1i}}$, $\mathbf{q_2'}\bar{\mathbf{Y}} = \dfrac{\sum_{i=1}^{m_2} q_{2i}Y_i}{\sum_{i=1}^{m_2} q_{2i}}$

(ii) $\mathbf{r_i'}\bar{\mathbf{C}}_\mathbf{i} = \dfrac{\sum_{j=1}^{n} r_{ij}C_{ij}}{\sum_{j=1}^{n} r_{ij}}$, $i \in [2]$

Now under H_0,

(i) $\mathbb{E}[\mathbf{q_1'}\bar{\mathbf{X}} - \mathbf{q_2'}\bar{\mathbf{Y}}] = 0$

(ii) $\text{Var}(\mathbf{q_1'}\bar{\mathbf{X}} - \mathbf{q_2'}\bar{\mathbf{Y}}) = \left[\dfrac{\sum_{i=1}^{m_1} q_{1i}^2}{(\sum_{i=1}^{m_1} q_{1i})^2} + \dfrac{\sum_{i=1}^{m_2} q_{2i}^2}{(\sum_{i=1}^{m_2} q_{2i})^2}\right]\sigma^2$

(iii) $\mathbb{E}[\mathbf{r_1'}\bar{\mathbf{C}}_\mathbf{1} - \mathbf{r_2'}\bar{\mathbf{C}}_\mathbf{2}] = 0$

(iv) $\text{Var}(\mathbf{r_1'}\bar{\mathbf{C}}_\mathbf{1} - \mathbf{r_2}\bar{\mathbf{C}}_\mathbf{2}) = \left[\dfrac{\sum_{i=1}^{n} r_{1i}^2}{(\sum_{i=1}^{n} r_{1i})^2} + \dfrac{\sum_{i=1}^{n} r_{2i}^2}{(\sum_{i=1}^{n} r_{2i})^2} - 2\dfrac{\sum_{i=1}^{n} r_{1i}r_{2i}\rho}{(\sum_{i=1}^{n} r_{1i})(\sum_{i=1}^{n} r_{2i})}\right]\sigma^2$

Now if we have to use Lyapunov's CLT, we have to find μ_i and σ_i for $Z_i = r_{1i}C_{1i} - r_{2i}C_{2i}$. Thus, $\mu_i = (r_{1i} - r_{2i})\mu$ and $\sigma_i^2 = (r_{1i}^2 + r_{2i}^2 - 2r_{1i}r_{2i}\rho)\sigma^2$. Thus,

$$s_n^2 = \sum_{i=1}^{n} \sigma_i^2 = \sigma^2 \sum_{i=1}^{n} (r_{1i}^2 + r_{2i}^2 - 2r_{1i}r_{2i}\rho)$$

From here, Lyapunov's Condition could be validated in certain cases. Otherwise we might treat the variables as continuous and do our analysis as in the previous case.

7 Suggestions for Improvement

We have taken the usual average for calculation of the unemployment rates and for taking the differences afterwards. However, as mentioned before, in a stratified sampling scheme, certain multipliers need to be used in order to give better estimates. However, due to the lack of computational power (precisely, for extracting data by blocks and sub-blocks and matching them), the method based on weighted averages could not be implemented. However, the procedure is exactly similar as mentioned before, and either the continuous counterpart of the statistic or *Lyapunov's CLT* could be used to derive the distribution of a similar test statistic.

8 Detailed Tables

Please check the tables appended at the end.

References

[1] N. S. Office, "Instructions to Field Staff *Vol. I*," Ministry of Statistics and Programme Implementation, Government of India, 2016.

[2] ——, "Periodic Labour Force Survey-*Quarterly Bulletin*, July-September 2020," Ministry of Statistics and Programme Implementation, Government of India, 2020.

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[4] A. W. Marshall and I. Olkin, "A *Family of Bivariate Distributions Generated by the Bivariate Bernoulli Distribution*," in the Journal of American Statistical Association, Vol.80, No. 390 332–338 (1985). https://doi.org/10.2307/2287890.

Comparison on the basis of gender: Male											
Sl. No.	State	Unemployment Rate				Significance between quarters					
		Q1	Q2	Q3	Q4	Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
						T1	p1	T2	p2	T3	p3
1	Jammu & Kashmir	6.0	8.7	10.4	11.5	-4.5540	**0.0000**	-0.7091	0.4783	-0.8888	0.3741
2	Himachal Pradesh	9.1	7.1	12.8	16.0	-0.2044	0.8381	-0.0326	0.9740	-3.5801	0.0003
3	Punjab	8.6	8.0	8.2	13.1	0.3883	0.6978	1.2745	0.2025	-6.4146	**0.0001**
4	Chandigarh	*	*	*	*	2.0257	0.0428	0.4448	0.6565	-3.4380	0.0006
5	Uttarakhand	7.3	76.0	9.2	27.6	-0.2577	0.7967	-3.3015	0.0010	-13.2356	**0.0001**
6	Haryana	7.5	6.8	6.9	13.2	0.9736	0.3302	0.5321	0.5947	-6.6764	**0.0001**
7	Delhi	11.6	9.8	12.9	17.5	-0.3917	0.6953	-1.2489	0.2117	-4.3307	**0.0001**
8	Rajasthan	9.4	10.6	13.4	26.4	-0.6544	0.5129	-2.1892	0.0286	-12.8941	**0.0001**
9	Uttar Pradesh	11.1	8.7	9.8	17.2	3.7499	**0.0002**	-3.0440	0.0023	-13.2574	**0.0001**
10	Bihar	8.5	8.4	9.7	12.7	0.2899	0.7719	-2.1145	0.0345	-4.1842	**0.0001**
11	Sikkim	*	*	*	*	-2.7871	0.0053	-0.4310	0.6664	-1.0241	0.3058
12	Arunachal Pradesh	*	*	*	*	2.2318	0.0256	-1.7964	0.0724	-1.4707	0.1414
13	Nagaland	*	*	*	*	-0.3857	0.6997	-2.6815	0.0073	-0.1887	0.8503
14	Manipur	*	*	*	*	-0.0218	0.9826	-1.7140	0.0865	-11.5253	**0.0001**
15	Mizoram	*	*	*	*	1.1052	0.2691	-0.4857	0.6272	0.9126	0.3614
16	Tripura	*	*	*	*	1.7361	0.0825	-1.0099	0.3126	-5.3563	**0.0001**
17	Meghalaya	*	*	*	*	1.6306	0.1030	*	*	-1.0734	0.2831
18	Assam	8.4	7.3	7.2	14.4	1.5275	0.1266	-1.4699	0.1416	-7.3994	**0.0001**
19	West Bengal	6.7	5.3	7.1	17.9	1.6539	0.0982	-1.9728	0.0485	-15.1826	**0.0001**
20	Jharkhand	8.3	8.1	11.9	33.4	0.0924	0.9263	-3.6790	**0.0002**	-15.8992	**0.0001**
21	Odisha	9.4	9.8	9.4	23.3	-0.6280	0.5300	-0.2653	0.7908	-9.4900	**0.0001**
22	Chhattisgarh	8.8	9.9	8.4	16.1	0.1145	0.9089	0.5582	0.5767	-6.5254	**0.0001**
23	Madhya Pradesh	12.5	11.4	12.7	29.8	0.9857	0.3243	-1.1391	0.2547	-16.8503	**0.0001**
24	Gujarat	4.6	3.4	3.6	9.6	0.9865	0.3239	0.1250	0.9005	-5.1617	**0.0001**
25	Daman & Diu	*	*	*	*	0.2931	0.7695	0.1292	0.8972	1.1743	0.2403
26	D & N Haveli	*	*	*	*	-0.8157	0.4146	0.0506	0.9596	*	*
27	Maharashtra	5.7	5.2	5.7	35.1	0.8295	0.4068	-2.6643	0.0077	-40.7620	**0.0001**
28	Andhra Pradesh	7.9	8.6	11.6	27.4	0.2634	0.7922	-2.7771	0.0055	-13.8451	**0.0001**
29	Karnataka	3.7	3.7	4.0	10.4	-0.5398	0.5893	-1.6555	0.0978	-14.8635	**0.0001**
30	Goa	*	*	*	*	-1.3837	0.1664	0.9217	0.3567	-3.7960	**0.0001**
31	Lakshadweep	*	*	*	*	*	*	*	*	*	*
32	Kerala	10.8	9.0	14.6	26.5	2.6119	0.0090	-7.4736	**0.0000**	-11.6214	**0.0001**
33	Tamil Nadu	6.8	7.0	8.0	18.2	-0.4842	0.6283	-1.9054	0.0567	-16.5069	**0.0001**
34	Puducherry	*	*	*	*	1.2631	0.2066	-1.7370	0.0824	-6.4738	**0.0001**
35	A & N Islands	*	*	*	*	1.7523	0.0797	0.4793	0.6317	-5.1460	**0.0001**
36	Telangana	9.0	10.2	11.4	24.7	-0.0369	0.9706	-2.6845	0.0073	-12.9057	**0.0001**

Comparison on the basis of gender: Female											
Sl. No.	State	Unemployment Rate				Significance between quarters					
		Q1	Q2	Q3	Q4	Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
						T1	p1	T2	p2	T3	p3
1	Jammu & Kashmir	28.9	36.1	36.8	37.9	-3.1722	0.0015	-2.4603	0.0139	0.9754	0.3294
2	Himachal Pradesh	15.9	8.6	12.9	12.6	1.5965	0.1104	-1.6081	0.1078	-0.6649	0.5061
3	Punjab	14.3	16.9	14.2	14.3	-2.0937	0.0363	0.1458	0.8841	0.7465	0.4554
4	Chandigarh	*	*	*	*	0.6309	0.5281	-1.1414	0.2537	-1.3332	0.1825
5	Uttarakhand	23.1	16.9	11.1	22.9	1.2092	0.2266	1.7225	0.0850	-2.0320	0.0422
6	Haryana	5.3	6.8	7.6	14.0	-1.3114	0.1897	-0.0758	0.9396	-2.8587	0.0043
7	Delhi	12.1	12.1	12.7	20.4	-1.0445	0.2963	-1.0199	0.3078	-0.4712	0.6375
8	Rajasthan	11.2	9.4	9.4	14.2	0.3466	0.7289	-0.7806	0.4350	-0.1847	0.8535
9	Uttar Pradesh	11.3	9.0	9.0	13.4	2.4395	0.0147	-1.9873	0.0469	-3.3280	0.0009
10	Bihar	10.3	10.8	12.3	14.2	-0.7212	0.4708	-1.3182	0.1874	-0.0900	0.9283
11	Sikkim	*	*	*	*	0.7146	0.4749	*	*	*	*
12	Arunachal Pradesh	*	*	*	*	1.7228	0.0849	0.4797	0.6314	-0.8469	0.3970
13	Nagaland	*	*	*	*	1.5662	0.1173	-1.0453	0.2959	-0.8664	0.3863
14	Manipur	*	*	*	*	0.0102	0.9918	-0.0362	0.9712	-6.8156	**0.0000**
15	Mizoram	*	*	*	*	0.2507	0.8020	0.2316	0.8168	0.7611	0.4466
16	Tripura	*	*	*	*	1.8558	0.0635	1.3713	0.1703	-1.9833	0.0473
17	Meghalaya	*	*	*	*	-1.2952	0.1953	-0.2294	0.8185	-0.7833	0.4335
18	Assam	18.9	20.1	19.1	20.9	-2.8980	0.0038	-0.1773	0.8593	-0.4034	0.6866
19	West Bengal	4.3	5.9	6.9	12.0	-3.1400	0.0017	-0.4253	0.6706	-3.6767	**0.0002**
20	Jharkhand	8.1	4.8	10.9	24.4	1.0629	0.2878	-2.0192	0.0435	-5.1465	**0.0000**
21	Odisha	16.8	14.6	18.2	21.5	0.4280	0.6687	-1.1661	0.2436	-1.8685	0.0617
22	Chhattisgarh	9.6	8.8	12.1	12.7	0.1504	0.8805	-2.1077	0.0351	-0.1728	0.8628
23	Madhya Pradesh	10.0	12.0	11.8	24.8	-1.0281	0.3039	0.3341	0.7383	-4.1230	**0.0000**
24	Gujarat	3.0	2.8	4.2	14.4	0.3630	0.7166	-1.3290	0.1839	-0.6713	0.5020
25	Daman & Diu	*	*	*	*	*	*	*	*	*	*
26	D & N Haveli	*	*	*	*	*	*	*	*	*	*
27	Maharashtra	6.4	6.6	8.0	37.2	0.0115	0.9908	-1.9816	0.0475	-15.5851	**0.0000**
28	Andhra Pradesh	15.8	13.6	15.2	22.0	1.8847	0.0595	-1.9234	0.0544	-2.4996	0.0124
29	Karnataka	4.9	6.3	5.2	21.0	-1.9575	0.0503	0.4619	0.6442	-4.4894	**0.0000**
30	Goa	*	*	*	*	-0.9390	0.3477	-1.4257	0.1540	-0.4528	0.6507
31	Lakshadweep	*	*	*	*	*	*	*	*	*	*
32	Kerala	20.7	19.9	20.4	29.4	0.3274	0.7434	-0.1843	0.8538	-2.5329	0.0113
33	Tamil Nadu	7.1	8.3	9.9	15.4	-0.9968	0.3189	-2.2218	0.0263	-3.1760	0.0015
34	Puducherry	*	*	*	*	-0.0004	0.9997	0.4695	0.6387	-2.4830	0.0130
35	A & N Islands	*	*	*	*	-1.1505	0.2499	-0.2500	0.8026	-1.6384	0.1013
36	Telangana	16.3	15.3	14.2	28.1	0.8365	0.4029	0.0917	0.9270	-5.9890	**0.0000**

Comparison on the basis of Age: 15 - 29 years											
Sl. No.	State	Unemployment Rate				Significance between quarters					
		Q1	Q2	Q3	Q4	Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
						T1	p1	T2	p2	T3	p3
1	**Jammu & Kashmir**	27.9	33.8	37.6	39.6	-2.1149	0.0344	-2.6958	0.0070	-1.7848	0.0743
2	**Himachal Pradesh**	23.2	15.6	31.6	33.9	0.2203	0.8256	-1.4159	0.1568	-1.6812	0.0927
3	**Punjab**	23.0	23.9	21.2	25.9	-1.4526	0.1463	1.7669	0.0772	-1.3944	0.1632
4	**Chandigarh**	*	*	*	*	2.5210	0.0117	-1.0712	0.2841	-2.4625	0.0138
5	**Uttarakhand**	19.7	20.7	21.5	37.7	0.2980	0.7657	-1.8355	0.0664	-6.1412	**0.0001**
6	**Haryana**	14.7	14.5	14.9	22.5	0.2789	0.7803	0.0624	0.9503	-4.8155	**0.0001**
7	**Delhi**	26.6	23.8	28.9	35.1	-0.2837	0.7767	-2.2483	0.0246	-1.6451	0.0999
8	**Rajasthan**	23.4	25.7	28.2	40.6	-0.6656	0.5056	-0.3777	0.7056	-5.0523	**0.0001**
9	**Uttar Pradesh**	24.8	19.0	21.8	30.1	4.8678	**0.0001**	4.8678	**0.0001**	-7.5785	**0.0001**
10	**Bihar**	25.6	25.1	25.3	28.8	0.0992	0.9210	-1.5812	0.1138	-2.7418	0.0061
11	**Sikkim**	*	*	*	*	-1.5252	0.1272	0.6664	0.5051	-2.6089	0.0091
12	**Arunachal Pradesh**	*	*	*	*	3.2637	0.0011	-1.4449	0.1485	-0.6565	0.5115
13	**Nagaland**	*	*	*	*	1.3772	0.1684	-1.8478	0.0065	-0.8775	0.3802
14	**Manipur**	*	*	*	*	0.2174	0.8279	-0.2904	0.7715	-8.5457	**0.0001**
15	**Mizoram**	*	*	*	*	0.9615	0.3363	-0.0393	0.9687	0.5514	0.5814
16	**Tripura**	*	*	*	*	2.0806	0.0375	0.0490	0.9606	-2.4523	0.0142
17	**Meghalaya**	*	*	*	*	1.0554	0.2912	-1.8320	0.0670	-0.5939	0.5526
18	**Assam**	30.5	31.0	29.3	32.3	-1.2612	0.2072	-1.0377	0.2994	-1.4030	0.1606
19	**West Bengal**	15.7	13.5	18.5	31.2	0.0457	0.9635	-1.9292	0.0537	-6.0305	**0.0001**
20	**Jharkhand**	18.8	15.7	19.0	48.7	1.0865	0.2773	-2.4107	0.0159	-10.2221	**0.0001**
21	**Odisha**	28.1	28.3	26.3	40.1	-0.4714	0.6373	0.2121	0.8320	-4.7642	**0.0001**
22	**Chhattisgarh**	25.4	25.8	24.3	33.5	0.9210	0.3571	-0.7975	0.4252	-2.0592	0.0395
23	**Madhya Pradesh**	27.4	25.2	27.1	46.8	0.4001	0.6891	-0.4885	0.6252	-8.4456	**0.0001**
24	**Gujarat**	9.4	7.2	8.7	12.5	1.2888	0.1975	-0.4571	0.6476	-1.1844	0.2363
25	**Daman & Diu**	*	*	*	*	0.3288	0.7423	0.4526	0.6518	1.2015	0.2295
26	**D & N Haveli**	*	*	*	*	0.2862	0.7747	-0.6312	0.5279	-1.3790	0.1679
27	**Maharashtra**	13.9	12.9	13.8	46.3	1.1066	0.2685	-1.6613	0.0966	-21.8102	**0.0001**
28	**Andhra Pradesh**	31.2	29.5	33.1	46.4	2.2069	0.0273	-1.1161	0.2644	-5.2021	**0.0001**
29	**Karnataka**	10.3	11.6	10.6	30.1	-2.0082	0.0446	-0.3210	0.7482	-9.1773	**0.0001**
30	**Goa**	*	*	*	*	-2.5222	0.0117	-0.4947	0.6208	1.0886	0.2763
31	**Lakshadweep**	*	*	*	*	*	*	*	*	*	*
32	**Kerala**	38.2	36.3	40.5	49.7	0.3460	0.7294	-2.1746	0.0297	-3.2372	0.0021
33	**Tamil Nadu**	19.5	18.6	19.9	28.6	-0.3319	0.7400	-1.7822	0.0747	-5.6382	**0.0001**
34	**Puducherry**	*	*	*	*	0.6796	0.4967	-0.4715	**0.0001**	-3.6417	**0.0003**
35	**A & N Islands**	*	*	*	*	0.7114	0.4768	-0.3807	0.7035	-1.7581	0.0787
36	**Telangana**	27.4	27.1	28.8	42.3	0.6152	0.5384	-1.2492	0.2116	-7.1737	**0.0001**

Comparison on the basis of Age: > 15 years											
Sl. No.	State	Unemployment Rate				Significance between quarters					
		Q1	Q2	Q3	Q4	Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
						T1	p1	T2	p2	T3	p3
1	**Jammu & Kashmir**	11.0	15.4	17.2	18.1	-5.4999	**0.0001**	-2.0746	0.0380	-0.3160	0.7520
2	**Himachal Pradesh**	10.8	7.5	12.8	14.9	0.8753	0.3814	-0.9606	0.3367	-3.4054	0.0007
3	**Punjab**	9.7	9.6	9.3	13.1	-0.5013	0.6161	1.2019	0.2294	-5.3559	**0.0001**
4	**Chandigarh**	*	*	*	*	1.7645	0.0776	-0.1809	0.8565	-3.9249	**0.0001**
5	**Uttarakhand**	9.7	9.3	9.2	26.8	0.6010	0.5479	-1.4792	0.1391	-12.7811	**0.0001**
6	**Haryana**	7.1	6.8	7.0	13.2	0.2695	0.7875	0.4717	0.6372	-7.1731	**0.0001**
7	**Delhi**	11.7	10.1	12.9	17.9	-0.5347	0.5929	-1.6390	0.1012	-3.9445	**0.0001**
8	**Rajasthan**	9.7	10.4	12.7	24.6	-0.6237	0.5328	-2.3532	0.0186	-12.3595	**0.0001**
9	**Uttar Pradesh**	11.1	8.7	9.7	16.6	4.4513	**0.0001**	-3.5992	**0.0003**	-13.5461	**0.0001**
10	**Bihar**	8.6	8.6	10.0	12.9	0.1193	0.9050	-2.5541	0.0106	-3.9513	**0.0001**
11	**Sikkim**	*	*	*	*	-2.1939	0.0282	0.1508	0.8801	-1.6352	0.1020
12	**Arunachal Pradesh**	*	*	*	*	2.5791	0.0099	-1.0013	0.3167	-1.8018	0.0716
13	**Nagaland**	*	*	*	*	0.5416	0.5881	-2.1893	0.0286	-0.2150	0.8298
14	**Manipur**	*	*	*	*	0.1026	0.9183	-1.0496	0.2939	-12.4042	**0.0001**
15	**Mizoram**	*	*	*	*	1.0880	0.2766	0.1115	0.9112	1.0074	0.3137
16	**Tripura**	*	*	*	*	2.3973	0.0165	-0.2453	0.8063	-5.6914	**0.0001**
17	**Meghalaya**	*	*	*	*	0.1715	0.8638	-0.8064	0.4200	-1.0559	0.2910
18	**Assam**	10.3	9.9	9.6	15.6	-0.4388	0.6608	-1.2925	0.1962	-6.6158	**0.0001**
19	**West Bengal**	6.1	5.3	6.9	16.4	-0.0891	0.9290	-1.7920	0.0731	-15.1762	**0.0001**
20	**Jharkhand**	8.3	7.6	11.8	32.3	0.4612	0.6447	-3.9577	**0.0001**	-17.1889	**0.0001**
21	**Odisha**	11.0	10.8	11.3	22.9	-0.5985	0.5495	-0.6628	0.5075	-9.4856	**0.0001**
22	**Chhattisgarh**	9.0	9.7	9.3	16.1	0.4438	0.6572	-0.5883	0.5564	-5.5343	**0.0001**
23	**Madhya Pradesh**	12.1	11.5	12.5	28.9	0.4777	0.6329	-0.9497	0.3423	-17.3297	**0.0001**
24	**Gujarat**	4.3	3.3	3.6	10.3	0.8931	0.3718	-0.3346	0.7380	-4.7707	**0.0001**
25	**Daman & Diu**	*	*	*	*	0.3770	0.7061	0.4358	0.6630	1.1782	0.2387
26	**D & N Haveli**	*	*	*	*	-0.7215	0.4706	-0.1377	0.8905	-0.8415	0.4001
27	**Maharashtra**	5.8	5.5	6.2	35.5	0.6736	0.5006	-3.1251	0.0018	-44.6152	**0.0001**
28	**Andhra Pradesh**	10.0	9.9	12.6	26.0	1.5692	0.1166	-3.4612	0.0005	-13.3533	**0.0001**
29	**Karnataka**	3.8	4.3	4.3	13.0	-2.1306	0.0331	-1.0184	0.3085	-15.3131	**0.0001**
30	**Goa**	*	*	*	*	-1.7039	0.0884	-0.0569	0.9547	-3.1081	0.0019
31	**Lakshadweep**	*	*	*	*	*	*	*	*	*	*
32	**Kerala**	13.9	12.5	16.3	27.3	2.1087	0.0350	-5.5080	**0.0001**	-11.4599	**0.0001**
33	**Tamil Nadu**	6.8	7.4	8.6	17.3	-0.9885	0.3229	-2.7823	0.0054	-15.6147	**0.0001**
34	**Puducherry**	*	*	*	*	1.0601	0.2891	-1.5143	0.1299	-7.1055	**0.0001**
35	**A & N Islands**	*	*	*	*	0.7223	0.4701	0.0144	0.9885	-4.6432	**0.0001**
36	**Telangana**	10.9	11.4	12.1	25.5	0.5153	0.6063	-2.0927	0.0364	-14.7536	**0.0001**

Comparison on the basis of Age: All ages											
Sl. No.	State	Unemployment Rate				Significance between quarters					
		Q1	Q2	Q3	Q4	Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
						T1	p1	T2	p2	T3	p3
1	**Jammu & Kashmir**	11.0	15.5	17.3	18.2	-5.5057	**0.0001**	-2.2268	0.0260	-0.1331	0.8941
2	**Himachal Pradesh**	10.8	7.4	12.8	14.9	0.7169	0.4735	-0.9090	0.3633	-3.2698	0.0011
3	**Punjab**	9.8	9.8	9.4	13.3	-0.8151	0.4150	1.1687	0.2425	-5.3384	**0.0001**
4	**Chandigarh**	*	*	*	*	1.9770	0.0480	-0.0376	0.9700	-3.8208	**0.0001**
5	**Uttarakhand**	9.9	9.3	9.6	26.8	0.6194	0.5356	-1.7119	0.0869	-12.6143	**0.0001**
6	**Haryana**	7.1	6.8	7.0	13.3	0.3171	0.7512	0.3698	0.7115	-7.3288	**0.0001**
7	**Delhi**	11.7	10.2	12.9	17.9	-0.7181	0.4727	-1.5854	0.1129	-4.0341	**0.0001**
8	**Rajasthan**	9.7	10.4	12.7	24.6	-0.5494	0.5827	-2.3627	0.0181	-12.0502	**0.0001**
9	**Uttar Pradesh**	11.1	8.8	9.7	16.7	4.2975	**0.0001**	-3.5553	**0.0004**	-13.6946	**0.0001**
10	**Bihar**	8.6	8.6	10.0	12.8	0.0472	0.0266	-2.4320	0.8804	-3.9592	0.1066
11	**Sikkim**	*	*	*	*	-2.2171	0.9624	0.1505	0.0150	-1.6138	**0.0001**
12	**Arunachal Pradesh**	*	*	*	*	2.8312	0.0046	-1.0481	0.2946	-1.7642	0.0777
13	**Nagaland**	*	*	*	*	0.6543	0.5129	-2.7446	0.0061	-0.6602	0.5091
14	**Manipur**	*	*	*	*	-0.0379	0.9698	-1.3323	0.1828	-14.2741	**0.0001**
15	**Mizoram**	*	*	*	*	1.0943	0.2738	-0.0872	0.9305	1.0833	0.2787
16	**Tripura**	*	*	*	*	2.3887	0.0169	-0.1667	0.8676	-5.6982	**0.0001**
17	**Meghalaya**	*	*	*	*	0.1800	0.8572	-0.6718	0.5017	-1.1800	0.2380
18	**Assam**	10.2	9.8	9.6	15.6	-0.4431	0.6577	-1.2587	0.2081	-6.6465	**0.0001**
19	**West Bengal**	6.1	5.5	7.0	16.5	-0.1760	0.8603	-1.8541	0.0637	-15.1251	**0.0001**
20	**Jharkhand**	8.3	7.5	11.8	32.0	0.4101	0.6817	-4.0047	**0.0001**	-17.2343	**0.0001**
21	**Odisha**	10.9	10.8	11.3	22.9	-0.4835	0.6287	-0.6290	0.5293	-9.5331	**0.0001**
22	**Chhattisgarh**	9.0	9.7	9.3	16.1	0.3983	0.6904	-0.7128	0.4759	-5.4883	**0.0001**
23	**Madhya Pradesh**	12.1	11.5	12.5	28.9	0.3182	0.7503	-0.9096	0.3630	-17.3783	**0.0001**
24	**Gujarat**	4.3	3.3	3.7	10.5	1.0071	0.3139	-0.5167	0.6054	-4.9157	**0.0001**
25	**Daman & Diu**	*	*	*	*	0.3828	0.7018	0.4275	0.6690	1.1739	0.2404
26	**D & N Haveli**	*	*	*	*	-0.6514	0.5148	-0.1313	0.8955	-0.7401	0.4592
27	**Maharashtra**	5.8	5.5	6.3	35.6	0.6643	0.5065	-3.2619	0.0011	-44.6679	**0.0001**
28	**Andhra Pradesh**	10.1	10.0	12.6	26.0	1.5209	0.1283	-3.3465	0.0008	-13.2469	**0.0001**
29	**Karnataka**	3.9	4.3	4.3	13.0	-1.6367	0.1017	-1.0894	0.2760	-15.2827	**0.0001**
30	**Goa**	*	*	*	*	-1.6594	0.0970	-0.0564	0.9550	-3.1934	0.0014
31	**Lakshadweep**	*	*	*	*	*	*	*	*	*	*
32	**Kerala**	13.9	12.5	16.4	27.3	2.1206	0.0340	-5.8252	**0.0001**	-11.2444	**0.0001**
33	**Tamil Nadu**	6.9	7.4	8.6	17.4	-1.0130	0.3111	-2.8885	0.0039	-15.6348	**0.0001**
34	**Puducherry**	*	*	*	*	0.9470	0.3436	-1.4297	**0.1528**	-7.1232	**0.0001**
35	**A & N Islands**	*	*	*	*	0.7009	0.4834	0.1893	0.8499	-4.7553	**0.0001**
36	**Telangana**	10.9	11.5	12.1	25.5	0.5676	0.5703	-2.1327	0.0330	-14.8450	**0.0001**

Comparison on the basis of Education: Upto Diploma / High School							
Sl. No.	State	Significance between quarters					
		Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
		T1	p1	T2	p2	T3	p3
1	Jammu & Kashmir	-2.1149	0.0344	-2.6958	0.0070	-1.7848	0.0743
2	Himachal Pradesh	0.2203	0.8256	-1.4159	0.1568	-1.6812	0.0927
3	Punjab	-1.4526	0.1463	1.7669	0.0772	-1.3944	0.1632
4	Chandigarh	2.5210	0.0117	-1.0712	0.2841	-2.4625	0.0138
5	Uttarakhand	0.2980	0.7657	-1.8355	0.0664	-6.1412	**0.0001**
6	Haryana	0.2789	0.7803	0.0624	0.9503	-4.8155	**0.0001**
7	Delhi	-0.2837	0.7767	-2.2483	0.0246	-1.6451	0.0999
8	Rajasthan	-0.6656	0.5056	-0.3777	0.7056	-5.0523	**0.0001**
9	Uttar Pradesh	4.8678	**0.0001**	4.8678	**0.0001**	-7.5785	**0.0001**
10	Bihar	0.0992	0.9210	-1.5812	0.1138	-2.7418	**0.0061**
11	Sikkim	-1.5252	0.1272	0.6664	0.5051	-2.6089	**0.0091**
12	Arunachal Pradesh	3.2637	0.0011	-1.4449	0.1485	-0.6565	0.5115
13	Nagaland	1.3772	0.1684	-1.8478	0.0065	-0.8775	0.3802
14	Manipur	0.2174	0.8279	-0.2904	0.7715	-8.5457	**0.0001**
15	Mizoram	0.9615	0.3363	-0.0393	0.9687	0.5514	0.5814
16	Tripura	2.0806	0.0375	0.0490	0.9606	-2.4523	0.0142
17	Meghalaya	1.0554	0.2912	-1.8320	0.0670	-0.5939	0.5526
18	Assam	-1.2612	0.2072	-1.0377	0.2994	-1.4030	0.1606
19	West Bengal	0.0457	0.9635	-1.9292	0.0537	-6.0305	**0.0001**
20	Jharkhand	1.0865	0.2773	-2.4107	0.0159	-10.2221	**0.0001**
21	Odisha	-0.4714	0.6373	0.2121	0.8320	-4.7642	**0.0001**
22	Chhattisgarh	0.9210	0.3571	-0.7975	0.4252	-2.0592	0.0395
23	Madhya Pradesh	0.4001	0.6891	-0.4885	0.6252	-8.4456	**0.0001**
24	Gujarat	1.2888	0.1975	-0.4571	0.6476	-1.1844	0.2363
25	Daman & Diu	0.3288	0.7423	0.4526	0.6518	1.2015	0.2295
26	D & N Haveli	0.2862	0.7747	-0.6312	0.5279	-1.3790	0.1679
27	Maharashtra	1.1066	0.2685	-1.6613	0.0966	-21.8102	**0.0001**
28	Andhra Pradesh	2.2069	0.0273	-1.1161	0.2644	-5.2021	**0.0001**
29	Karnataka	-2.0082	0.0446	-0.3210	0.7482	-9.1773	**0.0001**
30	Goa	-2.5222	0.0117	-0.4947	0.6208	1.0886	0.2763
31	Lakshadweep	*	*	*	*	*	*
32	Kerala	0.3460	0.7294	-2.1746	0.0297	-3.2372	0.0021
33	Tamil Nadu	-0.3319	0.7400	-1.7822	0.0747	-5.6382	**0.0001**
34	Puducherry	0.6796	0.4967	-0.4715	**0.0001**	-3.6417	**0.0003**
35	A & N Islands	0.7114	0.4768	-0.3807	0.7035	-1.7581	0.0787
36	Telangana	0.6152	0.5384	-1.2492	0.2116	-7.1737	**0.0001**

Comparison on the basis of Education: Graduate / Post-graduate							
Sl. No.	State	Significance between quarters					
		Q1 vs Q2		Q2 vs Q3		Q3 vs Q4	
		T1	p1	T2	p2	T3	p3
1	Jammu & Kashmir	-3.1928	0.0014	-2.3055	0.0211	1.2912	0.1967
2	Himachal Pradesh	1.0062	0.3143	0.2551	0.7987	-1.3927	0.1637
3	Punjab	-3.6781	0.0002	0.9744	0.3299	0.7499	0.4533
4	Chandigarh	0.9164	0.3594	-0.6257	0.5315	0.0135	0.9892
5	Uttarakhand	-1.3502	0.1769	1.7027	0.0886	-5.1563	**0.0001**
6	Haryana	-0.3757	0.7071	2.3192	0.0204	-2.5832	0.0098
7	Delhi	-0.2996	0.7645	-1.4431	0.1490	-0.3310	0.7406
8	Rajasthan	-0.8043	0.4212	0.0605	0.9517	-4.0196	**0.0001**
9	Uttar Pradesh	1.2489	0.2117	-1.2924	0.1962	-1.8658	0.0621
10	Bihar	0.1563	0.8758	-0.1856	0.8528	-2.1923	0.0284
11	Sikkim	-1.0529	0.2924	0.7954	0.4264	-0.7976	0.4251
12	Arunachal Pradesh	3.1755	0.0015	0.1960	0.8446	-0.4384	0.6611
13	Nagaland	1.2999	0.1936	-1.8885	0.0590	0.0143	0.9886
14	Manipur	-0.0446	0.9645	-0.9427	0.3458	-4.1377	**0.0001**
15	Mizoram	1.2815	0.2000	-0.8712	0.3836	-0.4606	0.6451
16	Tripura	2.1395	0.0324	-0.6628	0.5075	-1.8513	0.0641
17	Meghalaya	1.2200	0.2225	-0.1568	0.8754	-0.6769	0.4985
18	Assam	-0.8813	0.3782	-0.7708	0.4408	-1.9188	0.0550
19	West Bengal	-1.9494	0.0513	0.6989	0.4846	-3.9412	**0.0001**
20	Jharkhand	0.3447	0.7303	-1.7441	0.0811	-4.7560	**0.0001**
21	Odisha	-2.2714	0.0231	-0.1640	0.8698	-4.0272	**0.0001**
22	Chhattisgarh	0.1048	0.9165	-1.5607	0.1186	0.6339	0.5262
23	Madhya Pradesh	-0.3901	0.6965	0.7874	0.4310	-4.4189	**0.0001**
24	Gujarat	2.9727	0.0030	-0.4221	0.6730	0.3021	0.7625
25	Daman & Diu	0.0000	1.0000	0.0000	1.0000	0.7077	0.4791
26	D & N Haveli	-0.4363	0.6626	-0.3911	0.6957	-1.1371	0.2555
27	Maharashtra	0.6170	0.5372	-0.2392	0.8109	-17.1745	**0.0001**
28	Andhra Pradesh	1.3564	0.1750	0.0799	0.9364	-2.7300	0.0063
29	Karnataka	-2.2611	0.0238	-0.6820	0.4952	-2.5339	0.0113
30	Goa	-1.2677	0.2049	-0.1644	0.8694	0.1180	0.9061
31	Lakshadweep	*	*	*	*	*	*
32	Kerala	0.0582	0.9536	-1.6629	0.0963	-1.1382	0.2551
33	Tamil Nadu	0.1041	0.9171	0.5930	0.5532	-3.4854	0.0005
34	Puducherry	0.9225	0.3563	0.4808	0.6307	-2.2020	0.0277
35	A & N Islands	-0.0169	0.9865	0.1699	0.8651	-0.8918	0.3725
36	Telangana	-0.6059	0.5446	0.0255	0.9796	-3.5049	0.0005